

Concepts
Steganography is a technique of hiding secret information inside a normal-looking medium such as an image, audio, video, or text

a) Relationship between the concepts

- * Data hiding is the process of concealing secret information inside a cover medium
- * watermarking embeds information such as ownership or copyright details into digital content.
- * Survivability ensures that the hidden information remains available even after compression, modification, or attacks
- * these three concepts together improve secure communication

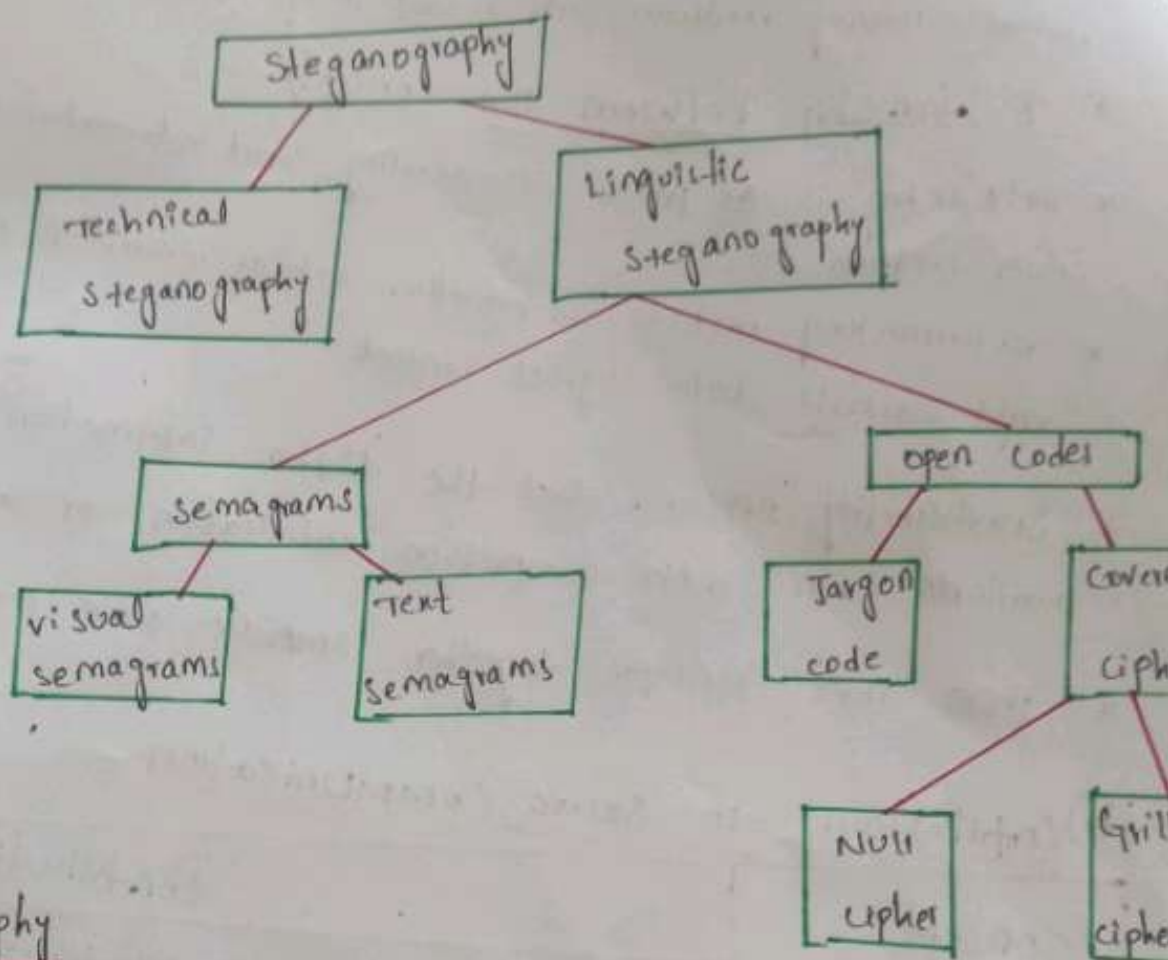
b) Contribution to Secure Communication

Concept	Contribution
Data hiding	Provides confidentiality by concealing the existence of secret information
Water marking	Provides ownership, authentication, and content protection
Survivability	Makes hidden information resistant to modification and attacks

Real - world Applications

Data hiding: Secret message hidden inside an image

Water marking: copy right water mark embedded in a digital



* Steganography

A technical of hiding secret information inside a normal-looking text or communication medium

* cover-based

Steganography that uses a hidden cover medium to hide secret information

* text

secret information is hidden inside a text document

* Image

secret information is hidden inside a digital image

this is one of the most common steganography methods.

important branch of mathematics used in cryptography. It provides the mathematical principles required to design secure encryption and decryption techniques. Concepts such as Modular Inverse, extended Euclid Algorithm, Fermat's Little theorem, Euler's phi function, and Euler's theorem are especially important in public-key cryptography.

a) Mathematical Connections Between the Concepts

① Extended Euclid Algorithm

The extended Euclidean Algorithm is an extension of the Euclidean algorithm. It is used to find the greatest common divisor (GCD) of two integers. More importantly, in cryptography, it can be used to find the modular inverse of a number.

if

$$\gcd(a, n) = 1$$

then the modular inverse of a modulo n exists. Then also algorithm finds integers x and y such that:

$$ax + ny = 1$$

therefore

$$ax \equiv 1 \pmod{n}$$

and x becomes the modular inverse of a.

2) Modular Inverse

The modular inverse of a modulo n is a number a^{-1} that:

$$a \times a^{-1} \equiv 1 \pmod{n}$$

the modular inverse exists only when a and n are

Number Theory Concepts used in Cryptography
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2) Modular Inverse

The Modular Inverse of a modulo n is a number a^{-1} such that:

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co prime meaning:

$$\gcd(a, n) = 1$$

The extended Euclid Algorithm is commonly used to calculate this inverse efficiently.

3) Euler's Phi Function

Euler's Phi Function, represented by $\phi(n)$, gives the number of positive integers less than or equal to n that are relatively prime to n .

For example:

$$\phi(10) = 4$$

because 1, 3, 7, and 9 are relatively prime to 10

For two prime numbers p and q :

$$\phi(pq) = (p-1)(q-1)$$

This property is very important in RSA cryptography

4) Euler's Theorem

Euler's theorem states that if

$$\gcd(a, n) = 1$$

then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Euler's theorem provides an important mathematical foundation for public-key cryptography algorithms

...ing the original message

Fermat's Little theorem

Fermat's Little theorem applies when the modulus is a Prime Number

If p is prime and a is not divisible by p , then:

$$a^{p-1} \equiv 1 \pmod{p}$$

It is closely related to Euler's theorem because for a prime number:

$$\phi(p) = p-1$$

Therefore, Fermat's little theorem can be considered a special case of Euler's theorem

Mathematical Relationship

The concepts can be connected as:

Euler's phi function $\rightarrow$ Euler's theorem $\rightarrow$ Fermat's little theorem and

Extended Euclid Algorithm $\leftrightarrow$ Modular Inverse

These concepts together provides mathematical tools used in cryptographic key generation and modular arithmetic

Role in Encryption and Decryption

Euler's phi function

Euler's phi function is particularly important in RSA key

$$n = p \times q$$

Euler's phi function



calculate $\phi(n)$



choose public exponent e



Extended Euclid Algorithm



Find Modular Inverse



calculate private key d



RSA Decryption

Conclusion:-

Number theory forms the mathematical foundation of Public key cryptography. Euler's phi function is essential for RSA key generation while Euler's theorem explains Important Modular Relationships. Fermat's Little theorem provides a special prime-number case of Euler's theorem.